

T3 matched EFT Lagrangian terms: EFT_1_after_F_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F, S1, S2. Remaining active T3 fields: none.

Field combinations 1 of 6

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(1)}\right)^2}{2\epsilon}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(1)}\right)^2}{2\epsilon}$		
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+C_{H2} H_{a_1} H_{a_1}^\dagger$		
								$-\overline{H_{a_1} \mu^2} H_{a_1}^\dagger$		
cont.								$+4\hbar\overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger$		
								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H1}^{(1)} \lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$		
								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H2}^{(1)} \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$		
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(1)}\right)^2}{2\epsilon}$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$	$-\frac{13\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(1)}\right)^2}{2\epsilon}$		
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+C_{H2} H_{a_1} H_{a_1}^\dagger$		
								$-\overline{H_{a_1} \mu^2} H_{a_1}^\dagger$		
cont.								$+4\hbar\overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger$		
								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H1}^{(1)} \lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$		
								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H2}^{(1)} \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$		
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{\hbar H_{a_3} H_{a_4} H_{a_1}^\dagger H_{a_2}^\dagger \mathcal{I}_{H1,a_1,a_3}{}^1 \mathcal{I}_{H1,a_2,a_4}{}^1 \left(\lambda_{H1}^{(1)}\right)^2}{2\epsilon}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$
								$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{2\epsilon}$		
cont.								$+CH21 H_{a_1} H_{a_1}^\dagger$		
								$-\overline{H_{a_1} \mu^2} H_{a_1}^\dagger$		
								$+2\hbar\overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger$		
								$+\hbar C_{H2} H_{a_2} H_{a_1}^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{2\epsilon}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$
								$+\frac{\hbar H_{a_3} H_{a_4} H_{a_1}^\dagger H_{a_2}^\dagger \mathcal{I}_{H2,a_1,a_3}{}^1 \mathcal{I}_{H2,a_2,a_4}{}^1 \left(\lambda_{H2}^{(1)}\right)^2}{2\epsilon}$		
cont.								$+\frac{+C_{H2} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{\sqrt{3}\epsilon}$		
								$+\frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$		
								$+CH21\hbar H_{a_2} H_{a_1}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{2\epsilon}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{5\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{6\epsilon}$
								$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{2\epsilon}$		
cont.								$+\frac{+C_{H2} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{\epsilon}$		
								$+\frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$		

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 H$	$S_1 B_{\mu\nu}$	$S_1 W_{\mu\nu}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_1}^{(1)}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
								$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_1}^{(2)}\right)^2$		
								$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H_1}^{(1)}\lambda_{H_1}^{(2)}}{\sqrt{3}}$		
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_1}^{(1)}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
								$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_1}^{(2)}\right)^2$		
								$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H_1}^{(1)}\lambda_{H_1}^{(2)}}{\sqrt{3}}$		
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H1,a_1,a_3}{}^1\mathcal{I}_{H1,a_2,a_4}{}^1\left(\lambda_{H_1}^{(1)}\right)^2$		
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_1}^{(2)}\right)^2$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_1}^{(2)}\right)^2$		$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_2 B_{\mu\nu}$	$S_2 W_{\mu\nu}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_2}^{(1)}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
								$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_2}^{(2)}\right)^2$		
								$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H_2}^{(1)}\lambda_{H_2}^{(2)}}{\sqrt{3}}$		
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_2}^{(1)}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
								$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_2}^{(2)}\right)^2$		
								$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H_2}^{(1)}\lambda_{H_2}^{(2)}}{\sqrt{3}}$		
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_2}^{(2)}\right)^2$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H2,a_1,a_3}\mathcal{I}_{H2,a_2,a_4}\left(\lambda_{H_2}^{(1)}\right)^2$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H_2}^{(2)}\right)^2$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F B_{\mu\nu}$	$F W_{\mu\nu}$	$H \ell e$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0			$+\frac{\hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8} \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8} \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$
cont.										
cont.										
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+\frac{\sqrt{3} \hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3} \hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3} \hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\sqrt{3} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$
cont.										
cont.										
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+\frac{\hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3} \hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8} \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\sqrt{3} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$
cont.										
cont.										
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+\frac{\sqrt{3} \hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $-\frac{1}{8} \frac{\sqrt{3} \hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F B_{\mu\nu}$	$F W_{\mu\nu}$	$H \ell e$
cont.									$-\frac{1}{8}\overline{\hbar y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{\hbar y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
									$-\frac{1}{8}\sqrt{3}\overline{\hbar y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\overline{\hbar y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
cont.									$-\frac{1}{8}\overline{\hbar y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{\hbar y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2$	$-\frac{1}{2}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{8}\sqrt{3}\overline{\hbar y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{\hbar y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$
cont.									$-\frac{1}{8}\sqrt{3}\overline{\hbar y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{\hbar y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
cont.									$-\frac{1}{8}\sqrt{3}\overline{\hbar y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{\hbar y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $-\frac{1}{8}\sqrt{3}\overline{\hbar y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	

Field combinations 5 of 6

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	ℓ	S_1	F	H	ℓ	e			
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	+	$+\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F^3}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
									$+\frac{4h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{2h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F^3}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
									$-\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_2}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$				
									$-\frac{4h\log\left(\frac{\mu^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{2h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_1}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$				
cont.																			
											$-\frac{4h\log\left(\frac{\mu^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{2h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{4\left(M_F^2-M_{S_1}^2\right)^2}$		
cont.												$-\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_2}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)^2}$				
												$+\frac{2h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{2\left(M_F^2-M_{S_1}^2\right)}$				
												$-\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_1}^2M_{S_2}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)^2}$				
												$+\frac{2h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_1}^2M_{S_2}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
cont.																			
											$-\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_2}^2M_{S_1}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$					
cont.												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)^2}$				
												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
cont.																			
											$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$					
cont.												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
												$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$				
cont.																			
											$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$			$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}}{2\left(M_F^2-M_{S_1}^2\right)}$					
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	+	$+\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F^3}{3\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
									$+\frac{4h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{2h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F^3}{3\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
									$-\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_2}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{3\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
									$-\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$			$-\frac{2h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}C P_R \overline{L_{a_4,i_2}}^r M_{S_1}^2M_{S_2}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{3\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$			$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$				

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 H$	$S_1 S_2 F H \ell e$	$S_1 F H \ell e$
										$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}M_{S_1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\left(M_F^2-M_{S_1}^2\right)^2}$
cont.										$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}M_{S_1}^4y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\left(M_F^2-M_{S_1}^2\right)^2}$
										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S_1}^2\right)}$
										$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{2\left(M_F^2-M_{S_1}^2\right)}$
cont.										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{4\left(M_F^2-M_{S_1}^2\right)^2}$
										$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{4\left(M_F^2-M_{S_1}^2\right)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $+\frac{2h\log\left(\frac{\mu^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{2h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$	$-\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F^2}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F^2}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}L_{a_3,i_1}CP_RL_{a_4,i_2}{}^TM_{S_1}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$	$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$ $-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^4}{4\left(M_F^2-M_{S_1}^2\right)^2}$ $+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$
cont.								$+\frac{2h\log\left(\frac{\mu^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$	$+\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}L_{a_3,i_1}CP_RL_{a_4,i_2}{}^TM_{S_1}^2y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}M_{S_1}^2M_{S_2}^2\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}M_F}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$	$+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$ $+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{4\left(M_F^2-M_{S_1}^2\right)^2}$ $+\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}L_{a_1,i_2}PR e_{i_1}H_{a_1}M_{S_1}^2y_{1,i_2,r_1}Y_{e,i_3,i_1}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)^2}$
cont.										$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$
										$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger M_F^2}{2\left(M_F^2-M_{S_1}^2\right)}$
cont.										$-\frac{\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}PL_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}^\dagger M_F^2}{2\left(M_F^2-M_{S_1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	S_2	F	H	ℓ	S_1	F	H	ℓ	e
cont.																				
																$-\frac{\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}M_{S_1}^2y_{1,i_3,r_1}H_{a_1}{}^\dagger}{2\left(M_F^2-M_{S_1}^2\right)}$				
																$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\left(M_F^2-M_{S_1}^2\right)^2}$				
															$-\frac{\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}M_{S_1}^4y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\left(M_F^2-M_{S_1}^2\right)^2}$					

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F H \ell e$
								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_2,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$+\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$-\frac{h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1} M_F^4}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1} M_F^4}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $-\frac{h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{h \bar{y}_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_2,r_1} H_{a_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\sqrt{3} h \bar{y}_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F H \ell e$
cont.								$\frac{\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{\sqrt{3} h \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$\frac{\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
								$- \frac{\sqrt{3} h \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$\frac{\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$\frac{\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
								$- \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$+ \frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
								$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$+ \frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$+ \frac{h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$+ \frac{h \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$
cont.								$- \frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
cont.								$- \frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F H \ell e$
<i>cont.</i>								$\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2 S_2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)} - \frac{\sqrt{3} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2 S_2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2} + \frac{\sqrt{3} h y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
<i>cont.</i>								$+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)} + \frac{\sqrt{3} h y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
<i>cont.</i>								$+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2} + \frac{\sqrt{3} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
								$+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
<i>cont.</i>								$+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)} + \frac{\sqrt{3} h y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{\sqrt{3} h y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$
<i>cont.</i>								$- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2 \left(M_F^2 - M_{S_2}^2 \right)} - \frac{\sqrt{3} h y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{\sqrt{3} h y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
<i>cont.</i>								$- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2} - \frac{\sqrt{3} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
								$- \frac{\sqrt{3} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
<i>cont.</i>								$- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_{S_2}^2} \right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)} - \frac{\sqrt{3} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$
								$- \frac{\sqrt{3} h y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2}} \bar{e}_{i_2} P_L L_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$